

MA269 - Mock | Model Solutions

① a) i) Dimensions: $m = kg$ $v = ms^{-1}$ $t = s$ $g = ms^{-2}$
 $m_0 = kg$ $V = ms^{-1}$ $k = s^{-1}$ ✓ (from equ)

Choose lengthscales. ✓ E.g. mass = m_0 , $T = \frac{1}{k}$, $D = gT^2 = \frac{g}{k^2}$ ✓

Nondimensional variables (denoted with a $\hat{\cdot}$):

$$\hat{t} = \frac{t}{T}, \quad m(\hat{t}) = m_0 \hat{m}(\hat{t}), \quad V = \frac{D}{T} \hat{V} = \frac{g}{k} \hat{V}, \quad v(\hat{t}) = \frac{g}{k} \hat{v}(\hat{t}) \quad \checkmark$$

$$\frac{d}{dt} = \frac{d\hat{t}}{dt} \frac{d}{d\hat{t}} = k \frac{d}{d\hat{t}} \quad \checkmark \quad \text{So equations become}$$

$$k \frac{d}{d\hat{t}} (m_0 \hat{m} - m_0) = -k (m_0 \hat{m} - m_0) \Rightarrow \frac{d}{d\hat{t}} (\hat{m} - 1) = -(\hat{m} - 1) \quad \checkmark$$

$$k \frac{d}{d\hat{t}} \left(m_0 \hat{m} \frac{g}{k} \hat{v} \right) = - \left(\frac{g}{k} \hat{V} - \frac{g}{k} \hat{v} \right) k \frac{d}{d\hat{t}} (m_0 \hat{m}) - m_0 \hat{m} g$$

$$\Rightarrow \frac{d}{d\hat{t}} (\hat{m} \hat{v}) = - (\hat{V} - \hat{v}) \frac{d\hat{m}}{d\hat{t}} - \hat{m} \quad \checkmark$$

At time $\hat{t} = 0$, $\hat{v} = 0$, $\hat{m} = \frac{M}{m_0} = \hat{M}$ ✓

[8]

ii) Dimensionless constants:

$$\hat{M} = \frac{M}{m_0} \quad \checkmark \quad \text{initial mass of rocket}$$

$$\hat{V} = \frac{kV}{g} \quad \checkmark \quad \text{speed of gas, rate of gas, or inverse of gravity.} \quad [2]$$

2)

b) Use successive approximation method:

$$x \simeq x_0, \quad x_0^2 (x_0^2 + x_0 + 1)^2 = 0$$

$$\Rightarrow x_0 = 0 \text{ (double root)}$$

$$\text{or } x_0 = \frac{-1 \pm \sqrt{1^2 - 4}}{2} = \text{complex.}$$

At next order, $x_0 = 0 + e_0$ (with e_0 small)

$$e_0^2 (\underbrace{e_0^2 + e_0 + 1}_{\text{smaller than 1}})^2 - \epsilon = 0 \Rightarrow e_0^2 = \epsilon, \text{ so } e_0 = \pm \sqrt{\epsilon}$$

At next order, $x_0 = 0 \pm \sqrt{\epsilon} + e_1$ (e_1 smaller than $\sqrt{\epsilon}$):

$$(\pm \sqrt{\epsilon} + e_1)^2 ((\pm \sqrt{\epsilon} + e_1)^2 + (\pm \sqrt{\epsilon} + e_1) + 1)^2 - \epsilon = 0$$

$$(\underbrace{\epsilon \pm 2\sqrt{\epsilon}e_1 + e_1^2}_{\text{smaller}}) \underbrace{\left(1 \pm \sqrt{\epsilon} + o(\sqrt{\epsilon})\right)^2}_{1 \pm 2\sqrt{\epsilon} + o(\sqrt{\epsilon})} - \epsilon = 0$$

$$\cancel{\epsilon} \pm 2\sqrt{\epsilon}e_1 \pm 2\epsilon\sqrt{\epsilon} - \cancel{\epsilon} = 0 \Rightarrow e_1 = -\epsilon$$

$$\text{So } x \sim \pm \sqrt{\epsilon} - \epsilon.$$

[10]

$$c) \quad \text{F.T.} \quad \frac{d}{dt} \mapsto i\omega \quad x(t) \mapsto i \frac{d}{d\omega}$$

$$\therefore 2(i\omega)^2 \tilde{y} + i \frac{d}{d\omega} (i\omega \tilde{y}) + \tilde{y} = 0$$

$$\Rightarrow \frac{d\tilde{y}}{d\omega} + 2\omega \tilde{y} = 0 \Rightarrow \frac{1}{\tilde{y}} \frac{d\tilde{y}}{d\omega} = -2\omega$$

$$\Rightarrow \log \tilde{y} = -\omega^2 + C \Rightarrow \tilde{y} = A e^{-\omega^2}$$

$$y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} A e^{-\omega^2} e^{i\omega t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} A e^{-(\omega - \frac{1}{2}it)^2 - \frac{1}{4}t^2} d\omega$$

$$= \frac{A}{2\sqrt{\pi}} e^{-\frac{1}{4}t^2} = e^{-t^2/4} \text{ by boundary condition.}$$

[10]

3)

d) either:

$$1 = \frac{1}{s}$$

$$t = \frac{1}{s^2}$$

$$g(t-b) = \hat{g}(s)e^{-sb}$$

$$\hat{f}(s) = \frac{1}{s^2} - \left(\frac{1}{s^2} + \frac{1}{s}\right)e^{-s} = t 1_{t>0} - \left[(x+1)1_{x>0}\right]_{x=t-1}$$

$$= t 1_{t>0} - t 1_{t>1} = t 1_{0<t<1} \quad [10]$$

or:

$$f(t) = \frac{1}{2\pi i} \int_{\alpha-i\infty}^{\alpha+i\infty} \frac{1}{s^2} e^{st} - \frac{(s+1)e^{-s}}{s^2} e^{st} ds$$

only singularity is at $s=0$ so just need $\alpha > 0$.

$$\textcircled{1} \quad \frac{1}{s^2} e^{st} = \frac{1}{s^2} \left(1 + st + \frac{1}{2}s^2 t^2 + \dots\right) = \frac{1}{s^2} + \frac{t}{s} + \frac{1}{2}s t^2 + \dots$$

Residue = t

$e^{st} \Rightarrow$ For $t < 0$, close in Right Half Plane to get zero.
For $t > 0$, close in Left Half Plane to get

$$\frac{1}{2\pi i} \times 2\pi i \times \text{Res} = t.$$

$$\textcircled{2} \quad \frac{1}{s^2} (s+1)e^{s(t-1)} = \frac{1}{s^2} (s+1) \left(1 + s(t-1) + \frac{1}{2}s^2(t-1)^2 + \dots\right)$$

$$= \frac{1}{s^2} + \frac{(t-1)+1}{s} + \left(\frac{1}{2}(t-1)^2 + (t-1)\right) + \dots$$

Residue = $(t-1)+1 = t$

$e^{s(t-1)} \Rightarrow$ For $t < 1$, close in Right Half Plane to get zero.
For $t > 1$, close in Left Half Plane to get

$$\frac{1}{2\pi i} \times 2\pi i \times \text{Res} = t.$$

$$f(t) = \textcircled{1} - \textcircled{2} = t 1_{t>0} - t 1_{t>1} = t 1_{0<t<1}. \quad [10]$$

4)

$$(2) \quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{\omega^2 + 1} e^{i(\omega^2 + \omega)t} d\omega \quad \text{as } t \rightarrow \infty,$$

So apply steepest descents. $\checkmark\checkmark$ $g(\omega) = i(\omega^2 + \omega)$ $\checkmark$

Saddle point: $g'(\omega_s) = i(2\omega_s + 1) = 0 \checkmark \Rightarrow \omega_s = -\frac{1}{2} \checkmark$

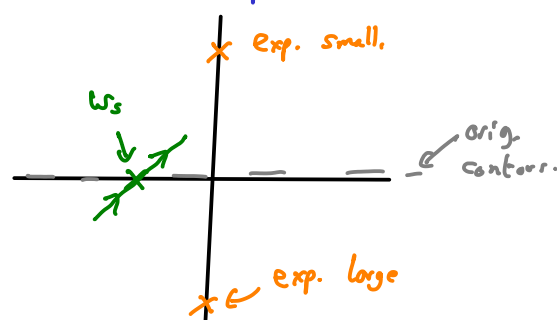
$$g''(\omega_s) = 2i \checkmark \quad g(\omega_s) = -i/4 \checkmark$$

So steepest descent contour goes through $\omega_s = -\frac{1}{2}$ in the direction $\sqrt{\frac{-2}{g''(\omega_s)}} = \pm e^{i\pi/4}$. Choose $\pm = +$. $\checkmark\checkmark$ (some notion of S.d. contour)

Integrand has poles at $\omega = \pm i$. In deforming onto the steepest descent contour, do we cross the poles? $\checkmark\checkmark$ (noticing poles)

It looks like we don't. But even if we did, they would contribute

$$O(e^{it((\pm i)^2 + \pm i)}) = O(e^{\mp t}),$$



So only the pole at $-i$ would give a contribution that would dominate the saddle point as $t \rightarrow \infty$, and we certainly don't cross that pole. $\checkmark\checkmark$ (for any argument about ignoring poles)

Can we deform onto steepest descent contour? At ∞ , provided $\text{Re}(i(\omega^2 + \omega)) < 0$ (which it is, as steepest descent contour goes downhill from $\text{Re}(i(\omega_s^2 + \omega_s)) = 0$), $|\text{integrand}| \leq \left| \frac{1}{\omega^2 + 1} \right|$,

$$\text{so at } |\omega| = R, \left| \int_{|\omega|=R} \dots d\omega \right| \leq \frac{2\pi R}{R^2 - 1} \rightarrow 0 \text{ as } R \rightarrow \infty. \text{ So OK.}$$

$\checkmark\checkmark$ (any argument about deforming contour at ∞)

$$\text{So, } f \sim \frac{1}{2\pi} \frac{e^{it((-\frac{1}{2})^2 - \frac{1}{2})}}{(-\frac{1}{2})^2 + 1} \sqrt{\frac{2\pi}{-t \cdot 2i}} \checkmark\checkmark \quad \text{(from front cover)}$$

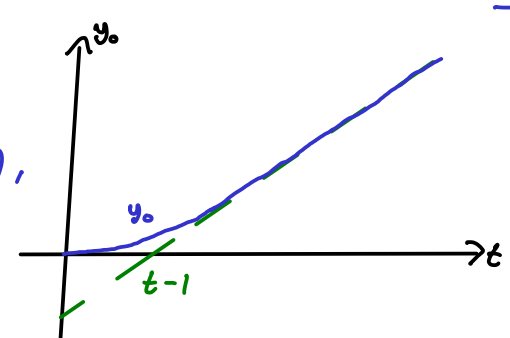
$$= \frac{2}{2\sqrt{\pi}t} \exp\left\{-\frac{i}{4}t + \frac{i\pi}{4}\right\} \checkmark\checkmark\checkmark$$

[20]

5)

③ a) To leading order, brs $\ddot{y}_0 + \dot{y}_0 = 1$ ✓, with $y_0(0) = \dot{y}_0(0) = 0$
 $\Rightarrow y_0 = \underbrace{t}_{\text{p.i.}} + \underbrace{A + Be^{-t}}_{\text{c.f.}} \checkmark \checkmark \text{ b.c.s} \Rightarrow y_0 = t + e^{-t} - 1 \checkmark$
[7]

As $t \rightarrow \infty$, $y_0 \sim t-1$, so $\dot{y}_0 = 1 = O(1)$,
 $\ddot{y}_0 \approx 0$, but ϵy_0 grows, so will
 become important when $t = O(\frac{1}{\epsilon})$.



b) Let $y = \epsilon^\beta \gamma$, $t = \epsilon^\alpha T$, $\frac{d}{dt} = \epsilon^{-\alpha} \frac{d}{dT}$

$$\epsilon^{\beta-2\alpha} \frac{d^2 \gamma}{dT^2} + \epsilon^{\beta-\alpha} \frac{d\gamma}{dT} + \epsilon^{1+\beta} \gamma = 1 \checkmark$$

We want new scaling to be the outer, so expect $\alpha < 0$.
 Then $\frac{d\gamma}{dT}$ term always dominates $\frac{d^2 \gamma}{dT^2}$, so aim to

make the $\frac{d^2 \gamma}{dT^2}$ term small and all others same size:

$$\beta - \alpha = 0, \quad 1 + \beta = 0 \Rightarrow \beta = -1, \quad \alpha = -1 \checkmark$$

$$\Rightarrow \epsilon \frac{d^2 \gamma}{dT^2} + \frac{d\gamma}{dT} + \gamma = 1 \checkmark$$

Try $\frac{d\gamma_0}{dT} + \gamma_0 = 1 \checkmark \Rightarrow \frac{1}{1-\gamma_0} \frac{d\gamma_0}{dT} = 1 \checkmark$

$$\Rightarrow \log(1-\gamma_0) = -T + C \Rightarrow \gamma_0 = 1 - Ae^{-T} \checkmark$$
[7]

c) Outer expansion of inner solution (with $t = T/\epsilon$) ✓
 $y_0(\frac{T}{\epsilon}) = \underbrace{\frac{T}{\epsilon}}_{\text{large}} - 1 + \underbrace{e^{-T/\epsilon}}_{\text{exponentially small.}} \checkmark$
Small

Inner expansion of outer solution with $(T = \epsilon t)$ ✓
 $\gamma_0(\epsilon t) = 1 - Ae^{-\epsilon t} = (1-A) + \underbrace{\epsilon At}_{\text{leading order}} - \underbrace{\frac{1}{2} A \epsilon^2 t^2}_{\text{2nd order}} + \dots$
Small ✓

6)

Matching $y_0(T/\varepsilon) \sim \frac{1}{\varepsilon} Y_0(\varepsilon t)$, and rewriting in terms of t :

$$\underset{\text{Large}}{t} - \underset{\text{small}}{1} \sim \underset{\text{too large}}{\frac{1-A}{\varepsilon}} + \underset{\text{large}}{At} - \underset{\text{small}}{\frac{1}{2} A \varepsilon t^2} \Rightarrow A=1 \quad [6]$$

d) Additive composite expansion:

$$y(t) \sim \underset{\text{inner}}{t-1+e^{-t}} + \underset{\text{outer}}{\frac{1}{\varepsilon}(1-e^{-\varepsilon t})} - t$$

$$= e^{-t} - 1 - \frac{1}{\varepsilon}(e^{-\varepsilon t} - 1)$$

outer-inner or inner-outer [3]

④

$$f(x) = 4 \int_{-\infty}^{\infty} g(t) f(x-t) dt = h(x) \quad \text{(noticing convolution)}$$

where $g(x) = 1_{x<0} e^{2x}$, $h(x) = e^{-2|x|}$

s. $\tilde{f}(k) = 4 \tilde{g}(k) \tilde{f}(k) = \tilde{h}(k)$ (convolution $\mapsto$ multiplication)

$$\Rightarrow \tilde{f}(k) = \frac{\tilde{h}(k)}{1-4\tilde{g}(k)}$$

$$\tilde{g}(k) = \int_{-\infty}^{\infty} 1_{x<0} e^{2x} e^{-ikx} dx = \int_{-\infty}^0 e^{(2-ik)x} dx$$

$$= \left[\frac{e^{(2-ik)x}}{2-ik} \right]_{-\infty}^0 = \frac{1}{2-ik}$$

$$\tilde{h}(k) = \int_{-\infty}^0 e^{2x} e^{-ikx} dx + \int_0^{\infty} e^{-2x} e^{-ikx} dx$$

$$= \left[\frac{e^{(2-ik)x}}{2-ik} \right]_{-\infty}^0 + \left[\frac{e^{-(2+ik)x}}{-(2+ik)} \right]_0^{\infty}$$

$$= \frac{1}{2-ik} + \frac{1}{2+ik} = \frac{4}{4+k^2}$$

7)

$$So \quad \hat{f}(k) = \frac{\frac{4}{4+k^2}}{1 - \frac{4}{2-ik}} = \frac{\frac{4}{(2+ik)(2-ik)}}{-\frac{2+ik}{2-ik}} = \frac{-4}{(2+ik)^2}$$

$$So \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{-4}{(2+ik)^2} e^{ikx} dk$$

only singularity is at $k = 2i$:

$$\frac{-4}{(2+ik)^2} e^{ikx} = \frac{4 e^{i(2i)x}}{(k-2i)^2} e^{i(k-2i)x} = \frac{4 e^{-2x}}{(k-2i)^2} + \frac{4 e^{-2x}}{(k-2i)} ix + 4e \frac{(ix)^2}{2} + \dots$$

$$Res = 4ix e^{-2x}$$

For $x < 0$, close contour in Lower Half Plane to get zero.

For $x > 0$, close contour in upper Half Plane to get.

$$f(x) = \frac{1}{2\pi} \cdot 2\pi i \cdot 4ix e^{-2x} = -4x e^{-2x}$$

$$So \quad f(x) = -4x e^{-2x} \quad 1_{x>0}.$$

⑤ Laplace, so $\widehat{\frac{df}{dt}} = -f(0) + s \hat{f}(s)$

$$\widehat{e^{at}} = \frac{1}{s-a}$$

$$\widehat{tf(t)} = -\frac{d\hat{f}}{ds}$$

$$\therefore \underbrace{-x e^{-x}}_{\widehat{\frac{dy}{dt}}} + s \hat{y} + \frac{\partial \hat{y}}{\partial s} = 0, \quad \hat{y}(0, s) = \frac{1}{(s+1)^2}$$

[4]

8)

$$\text{So } \frac{\partial \hat{y}}{\partial x} + s \hat{y} = x e^{-x} \Rightarrow \hat{y} = \underbrace{(ax+b)}_{\text{p.i.}} e^{-x} + \underbrace{A e^{-sx}}_{\text{c.f.}}$$

$$\begin{aligned} \text{differentiating, } \frac{\partial}{\partial x} \left((ax+b) e^{-x} \right) + s(ax+b) e^{-x} \\ = (a(s-1)x + a + (s-1)b) e^{-x} = (x+0) e^{-x} \\ \text{So } a = \frac{1}{s-1}, \quad b = \frac{-1}{(s-1)^2} \end{aligned}$$

$$\text{So } \hat{y} = \left(\frac{x}{s-1} - \frac{1}{(s-1)^2} \right) e^{-x} + A e^{-sx}$$

$$\text{@ } x=0, \quad \hat{y} = \frac{-1}{(s-1)^2} + A = \frac{1}{(s+1)^2} \Rightarrow A = \frac{1}{(s+1)^2} + \frac{1}{(s-1)^2} \quad [5]$$

$$\text{So } y(t) = \frac{1}{2\pi i} \int_{-i\infty+\alpha}^{i\infty+\alpha} \left[\left(\frac{x}{s-1} - \frac{1}{(s-1)^2} \right) e^{-x} + \left(\frac{1}{(s+1)^2} + \frac{1}{(s-1)^2} \right) e^{-sx} \right] e^{st} ds$$

either (by inspection):

$$\begin{aligned} y &= (x e^t - t e^t) e^{-x} 1_{t \geq 0} + \left[(s e^{-s} + s e^s) 1_{s > 0} \right]_{s=t-x} \\ &= (x e^t - t e^t) e^{-x} 1_{t \geq 0} + (t-x) \left(e^{x-t} + e^{t-x} \right) 1_{t > x} \quad [9] \\ &\quad \text{(continued later...)} \end{aligned}$$

or (by contour integration): (writing down inversion integral)

$$I_1 = \frac{1}{2\pi i} \int_{-i\infty+\alpha}^{i\infty+\alpha} \left(\frac{x}{s-1} - \frac{1}{(s-1)^2} \right) e^{-x} e^{st} ds$$

$$\begin{aligned} \text{pole at } s=1, \quad \text{integrand} &= \left(\frac{x}{s-1} - \frac{1}{(s-1)^2} \right) e^{-x} e^{st} \underbrace{\left(1 + (s-1)t + \frac{1}{2}(s-1)^2 t^2 + \dots \right)}_{e^{(s-1)t}} \\ &= -\frac{e^{t-x}}{(s-1)^2} + \frac{x-t}{s-1} e^{t-x} + (xt - \frac{1}{2}t^2) e^{t-x} + \dots \\ &\quad \uparrow \text{Residue} = (x-t) e^{t-x} \end{aligned}$$

So for $t < 0$, close in Right Half Plane to get $I_1 = 0$

for $t > 0$, close in Left Half Plane to get

$$I_1 = \frac{1}{2\pi i} \cdot 2\pi i (x-t) e^{t-x} = (x-t) e^{t-x} 1_{x > 0}.$$

9)

Similarly,
$$I_2 = \frac{1}{2\pi i} \int_{-i\infty+\alpha}^{i\infty+\alpha} \left(\frac{1}{(s+1)^2} + \frac{1}{(s-1)^2} \right) e^{-sx} e^{st} ds$$

Poles at $s=1$ and $s=-1$: At $s=1$,

$$\frac{e^{t-x}}{(s-1)^2} e^{(s-1)(t-x)} = \frac{e^{t-x}}{(s-1)^2} (1 + (s-1)(t-x) + \dots)$$

$$= \frac{e^{t-x}}{(s-1)^2} + \frac{(t-x)e^{t-x}}{s-1} + \dots \quad \text{Res} = (t-x)e^{t-x} \checkmark$$

At $s=-1$,
$$\frac{1}{(s+1)^2} e^{x-t} e^{(s+1)(t-x)} = \frac{e^{x-t}}{(s+1)^2} (1 + (s+1)(t-x) + \dots)$$

$$= \frac{e^{x-t}}{(s+1)^2} + \frac{(t-x)e^{x-t}}{s+1} + \dots \quad \text{Res} = (t-x)e^{x-t} \checkmark$$

For $t < x$, close in Right Half Plane to get $I_2 = 0$

For $t > x$, close in Left Half Plane to get $\checkmark$

$$I_2 = \frac{1}{2\pi i} \cdot 2\pi i \cdot \left(\underset{\text{Res: } s=-1}{(t-x)e^{t-x}} + \underset{\text{Res: } s=1}{(t-x)e^{x-t}} \right)$$

$$= (t-x)(e^{t-x} + e^{x-t}) \mathbb{1}_{t>x} \quad \checkmark$$

So $y = I_1 + I_2 = (x-t)e^{t-x} \mathbb{1}_{x>0} + (t-x)(e^{t-x} + e^{x-t}) \mathbb{1}_{t>x} \checkmark$
 (same result as first way. So ...) [9]

either way, get:

$$y = \begin{cases} (x-t)e^{t-x} & 0 < t < x \\ (t-x)e^{x-t} & 0 < x < t \end{cases} \checkmark \checkmark = |t-x| e^{-|t-x|} \quad [2]$$